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e-CAM121_CUONX



Lens Datasheet

Revision 1.0

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1 Introduction

e-CAM121_CUONX is a new mobile industry processor interface (MIPI) camera board from e-con Systems, a company with over two decades of experience in designing, developing, and manufacturing OEM cameras. e-CAM121_CUONX board interconnects 12.3 MP custom lens camera modules with a NVIDIA® Jetson Orin Nano™/Orin™ NX development kit. It has 1/2.3" optical form-factor, IMX412-AACK-D CMOS image sensor from Sony®. These 12.3 MP cameras are provided with a small form factor S-mount lens holder (also known as M12 board lens).

This document describes the optical specifications of the lens used in e-CAM121_CUONX.

2 Description

The lens used in e-CAM121_CUONX is a high resolution, 1/2.3" format, and wide-angle lens. It provides a maximum field of view (FOV) of 82° degrees with 1/2.3" format IMX412-AACK-D CMOS image sensor.

3 Optical Specifications

The optical specification of the lens is listed in the below table.

Description	Specification
Imager Format	1/2.3" (1/2.3" for IMX412)
Focal Length (mm)	4.3 mm (+/-5%)
Aperture (F/#)	2.8 (+/-5%)
FOV	82°(D), 72°(H), 56°(V) (with 1/2.3" IMX412)
Distortion	< 0.58%
Chief Ray Angle	<16.3°
Lens Barrel Thread	M12 x 0.5

Table 1: Optical Specifications of Lens



4 Mechanical Dimensions

The mechanical dimension of the lens used in the e-CAM121_CUONX is shown below.

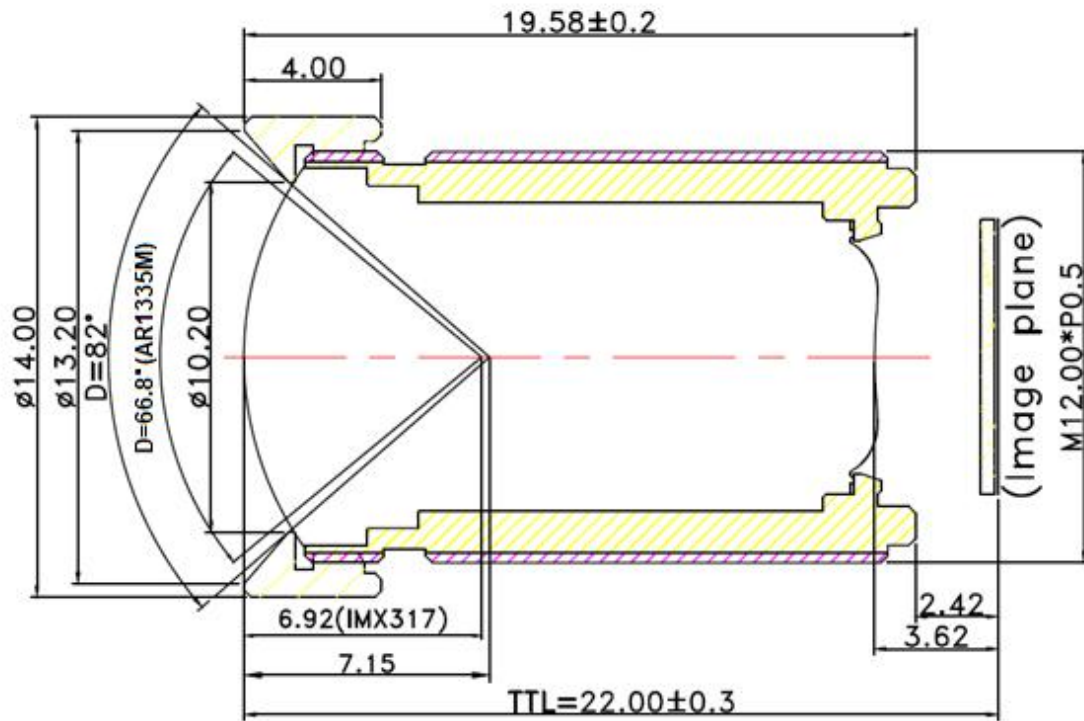


Figure 1: Lens Mechanical Dimensions

Note: All dimensions are in millimeter (mm).



Support

Contact Us

If you need any support on e-CAM121_CUONX product, please contact us using the Live Chat option available on our website - <https://www.e-consystems.com/>

Creating a Ticket

If you need to create a ticket for any type of issue, please visit the ticketing page on our website - <https://www.e-consystems.com/create-ticket.asp>

RMA

To know about our Return Material Authorization (RMA) policy, please visit the RMA Policy page on our website - <https://www.e-consystems.com/RMA-Policy.asp>

General Product Warranty Terms

To know about our General Product Warranty Terms, please visit the General Warranty Terms page on our website - <https://www.e-consystems.com/warranty.asp>



Revision History

Rev	Date	Description	Author
1.0	06-July-2023	Initial Draft	Products Marketing Team

